

2025-10-02 - Valentini (1)

Next lecture: [2025-10-03 - Valentini \(2\)](#)

Mean free path of particle $\sim 1 \text{ AU}$

only 1 collision coming to earth

Vlasov equation

when we integrate over velocity, we assumed thermodynamical equilibrium, but with little collision we don't have it

rarest lab density abt 10^{10} particles

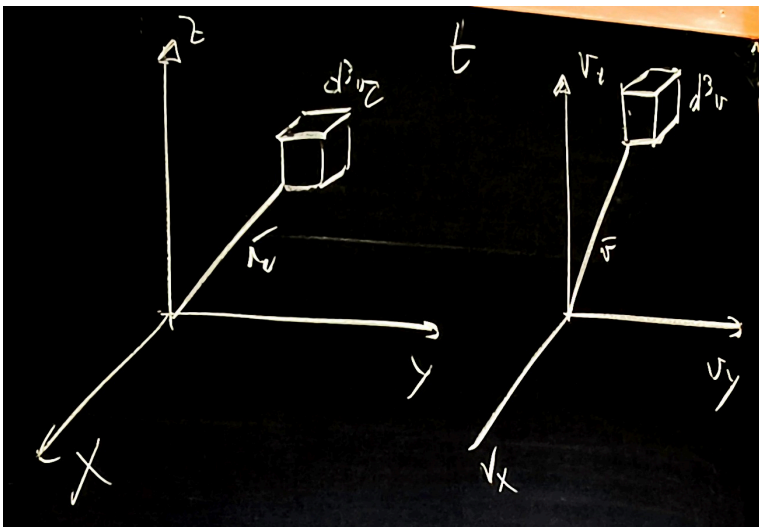
density sol wind abt 5 – 10 particles per cm^3

$f_\alpha \rightarrow$ distribution function

with α stands for ions (i) or electrons (e)

$$f_\alpha(\bar{r}, \bar{v}, t) d^3r d^3v = dN_\alpha \implies \int f_\alpha(\bar{r}, \bar{v}, t) d^3r d^3v = N_\alpha$$

pic 01 - phase space system of reference



Particle density

$$\int f_\alpha(\bar{r}, \bar{v}, t) d^3v = n_\alpha(\bar{r}, t)$$

Mean velocity of the gas

$$\frac{\int \bar{v} f_\alpha(\bar{r}, \bar{v}, t) d^3v}{n_\alpha(\bar{r}, t)} = \bar{\mu}_\alpha(\bar{r}, t)$$

Temperature (in non equilibrium we don't really know what it is, we call this in analogy with the equilibrium condition)

$$\frac{m_\alpha}{3} \frac{\int (\bar{v} - \bar{U}_\alpha) \cdot (\bar{v} - \bar{U}_\alpha) f_\alpha(\bar{r}, \bar{v}, t) d^3v}{n_\alpha(\bar{r}, t)} = T_\alpha(\bar{r}, t)$$

$$\left(\frac{3}{2}k_B T = \frac{1}{2}mv^2\right)$$

Pressure

$$m_\alpha \int (v_i - U_{i\alpha})(v_j - U_{j\alpha}) f_\alpha(\bar{r}, \bar{v}, t) d^3v = P_{ij}^\alpha(\bar{r}, t)$$

Charge

$$\frac{m_\alpha}{2} \int (\bar{v} - U_\alpha)^2 (\bar{v} - \bar{U}_\alpha) f_\alpha(\bar{r}, \bar{v}, t) d^3v = \bar{q}_\alpha = \rho_\alpha(\bar{r}, t)$$

Vlasov Equation

$$\frac{\partial f_\alpha}{\partial t} + \bar{v} \cdot \nabla f_\alpha + \frac{q_\alpha}{m_\alpha} \left[\bar{E} + \frac{\bar{v} \times \bar{B}}{c} \right] \cdot \nabla_v f_\alpha = \left. \frac{\partial f_\alpha}{\partial t} \right|_{\text{coll.}} = 0$$

CGS system

we couple with M.eq

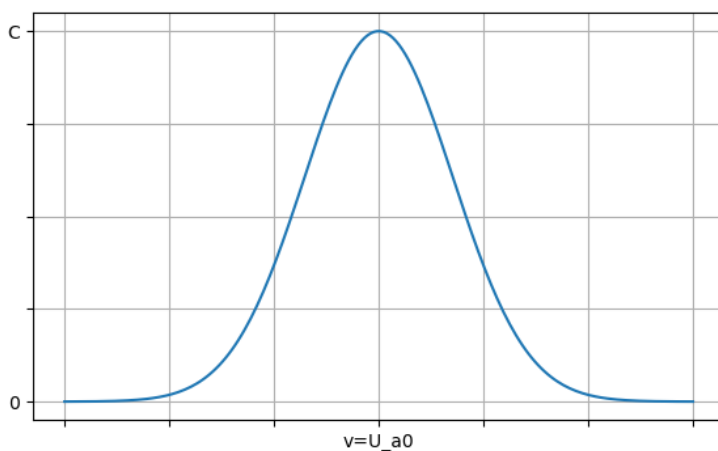
$$\left\{ \begin{array}{l} \nabla \cdot \bar{E} = 4\pi\rho = 4\pi \sum_\alpha q_\alpha \int f_\alpha d^3v \\ \nabla \cdot \bar{B} = 0 \\ \nabla \times \bar{E} = -\frac{1}{c} \frac{\partial \bar{B}}{\partial t} \\ \nabla \times \bar{B} = \frac{4\pi}{c} \sum_\alpha q_\alpha \underbrace{\int \bar{v} f_\alpha d^3v}_{\bar{J}_\alpha(\bar{r}, t)} + \frac{1}{c} \frac{\partial \bar{E}}{\partial t} \end{array} \right.$$

vlasov-maxwell equations

Vlasov maxwell distribution

$$f_{\alpha 0}(\bar{v}) = \frac{n_{\alpha 0} m_\alpha}{(2\pi k_B T_{\alpha 0})^{3/2}} \exp \left[-\frac{m_\alpha (\bar{v} - \bar{U}_{\alpha 0})^2}{2k_B T_\alpha} \right]$$

pic 02 - Vlasov maxwell distribution



verify at home

$$\int f_{\alpha 0}(\bar{r}, \bar{v}, t) d^3v = n_{\alpha 0}(\bar{r}, t)$$

$$\frac{\int \bar{v} f_{\alpha 0}(\bar{r}, \bar{v}, t) d^3v}{n_{\alpha 0}(\bar{r}, t)} = \bar{\mu}_{\alpha 0}(\bar{r}, t)$$

$$\frac{m_{\alpha 0}}{3} \frac{\int (\bar{v} - \bar{U}_{\alpha 0}) \cdot (\bar{v} - \bar{U}_{\alpha 0}) f_{\alpha 0}(\bar{r}, \bar{v}, t) d^3v}{n_{\alpha 0}(\bar{r}, t)} = T_{\alpha 0}(\bar{r}, t)$$

$$q_{\alpha 0} = 0$$

$$P_{ij}^{\alpha} = \mathcal{I}d \cdot (n_{\alpha 0} k_B T_{\alpha 0})$$

Then

- $f_{\alpha 0}(\bar{v})$ solution of vlasov eq
- $\frac{d}{dt} N_{\alpha} = 0$,
- $\mathcal{E}$ is a const of motion then $f(\mathcal{E})$ is solution of vlasov eq
- Entropy $S \equiv - \sum_{\alpha} f_{\alpha} \ln f_{\alpha}$ and $\frac{d}{dt} S = 0$
H theorem abt entropy, if you neglect collision total entropy is constant
- $I_n = \int f_{\alpha}^n d^3v d^3r$ Kaisimirs, if f_{α} is a sol of V.eq, $\frac{d}{dt} I_n = 0$
- if f_{α} is a sol of Vlasov eq, $g(f_{\alpha})$ is a sol of V.eq

Total derivative of f

$$\frac{d}{dt} f_{\alpha} = \frac{\partial f_{\alpha}}{\partial t} + \frac{\partial f_{\alpha}}{\partial \bar{r}} \frac{\partial \bar{r}}{\partial t} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial t}$$

we use *characteristics curves*

$$\frac{d}{dt} \bar{r} = \bar{v}, \quad \frac{d}{dt} \bar{v} = \frac{q_{\alpha}}{m_{\alpha}} \left(\bar{E} + \frac{\bar{v} \times \bar{B}}{c} \right)$$

and

$$\frac{d}{dt} f_{\alpha} = \frac{\partial f_{\alpha}}{\partial t} + \frac{\partial f_{\alpha}}{\partial \bar{r}} \bar{v} + \frac{\partial f}{\partial v} \frac{q_{\alpha}}{m_{\alpha}} \left(\bar{E} + \frac{\bar{v} \times \bar{B}}{c} \right) = 0$$

which is vlasov

$$\left. \frac{df_{\alpha}}{dt} \right|_{\text{orbit}} = 0$$

Two fluid equations

electrostatic approximation, unmagnetized plasma

$$\bar{B} = 0, \dot{\bar{B}} = 0$$

so

$$\begin{aligned}
\frac{\partial n_\alpha}{\partial t} + \nabla \cdot (n_\alpha \bar{U}_\alpha) &= 0 \\
m_\alpha n_\alpha \left[\frac{\partial U}{\partial t} + U_\alpha \cdot \nabla U_\alpha \right] &= -\nabla \cdot \overset{\leftrightarrow}{P}_\alpha + q_\alpha n_\alpha \bar{E} = -\nabla p_\alpha + q_\alpha n_\alpha (-\nabla \Phi) \\
\nabla \cdot \bar{E} &= 4\pi \sum_\alpha q_\alpha n_\alpha \\
\nabla \cdot \bar{B} &= 0 \\
\nabla \times \bar{E} &= -\frac{1}{c} \frac{\partial \bar{B}}{\partial t} = 0 \\
\nabla \times \bar{B} &= \frac{4\pi}{c} \sum_\alpha q_\alpha n_\alpha \bar{U}_\alpha + \frac{1}{c} \frac{\partial \bar{E}}{\partial t} = 0
\end{aligned}$$

in our case

$$-\nabla \cdot \overset{\leftrightarrow}{P}_\alpha \rightarrow -\nabla p_\alpha$$

with (ideal gas hypothesis)

$$p_\alpha = n_\alpha k_B T_\alpha$$

since

$$\nabla \times \bar{E} = 0 \implies \bar{E} = -\nabla \Phi$$

we eliminate one redundant equation

$$\begin{aligned}
\nabla \times \bar{B} &= \frac{4\pi}{c} \sum_\alpha q_\alpha n_\alpha \bar{U}_\alpha + \frac{1}{c} \frac{\partial \bar{E}}{\partial t} = 0 \\
\implies \nabla \cdot \left(\frac{4\pi}{c} \sum_\alpha q_\alpha n_\alpha \bar{U}_\alpha + \frac{1}{c} \frac{\partial \bar{E}}{\partial t} \right) &= 0 \\
\implies \frac{\partial}{\partial t} \nabla \cdot \bar{E} + 4\pi \sum_\alpha q_\alpha \nabla \cdot (n_\alpha \bar{U}_\alpha) &= 0
\end{aligned}$$

Now we use continuity equation and

$$\begin{aligned}
\implies \frac{\partial}{\partial t} \nabla \cdot \bar{E} + 4\pi \sum_\alpha q_\alpha \left(-\frac{\partial n_\alpha}{\partial t} \right) &= 0 \\
\implies \frac{\partial}{\partial t} \left(\nabla \cdot \bar{E} - 4\pi \sum_\alpha q_\alpha n_\alpha \right) &= 0 \\
\implies \nabla \cdot \bar{E} - 4\pi \sum_\alpha q_\alpha n_\alpha = K = 0
\end{aligned}$$

which does not have a time derivative and it's easier to use.

we say $K = 0$ its the only possible choice from poisson equation.

(eg. a 1D solution in a neutral space for $\nabla \cdot \bar{E} = K \neq 0$ is $\bar{E}(x) = Kx$, which diverges at infinity).

In the end we have the following set of equations

$$\frac{\partial n_\alpha}{\partial t} + \nabla \cdot (n_\alpha \bar{U}_\alpha) = 0$$

$$m_\alpha n_\alpha \left[\frac{\partial \bar{U}}{\partial t} + \bar{U}_\alpha \cdot \nabla \bar{U}_\alpha \right] = -\nabla p_\alpha + q_\alpha n_\alpha (-\nabla \Phi)$$

$$-\nabla^2 \Phi = 4\pi \sum_\alpha q_\alpha n_\alpha$$

$$p_\alpha = n_\alpha k_B T_\alpha$$

Next lecture: [2025-10-03 - Valentini \(2\)](#).